

System Architecture Document

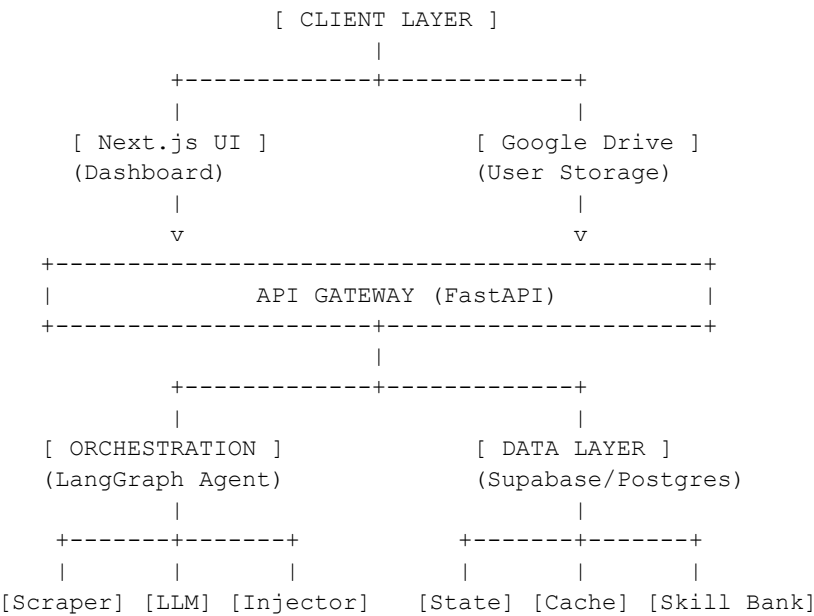
Project: CareerTailor (Production V1)

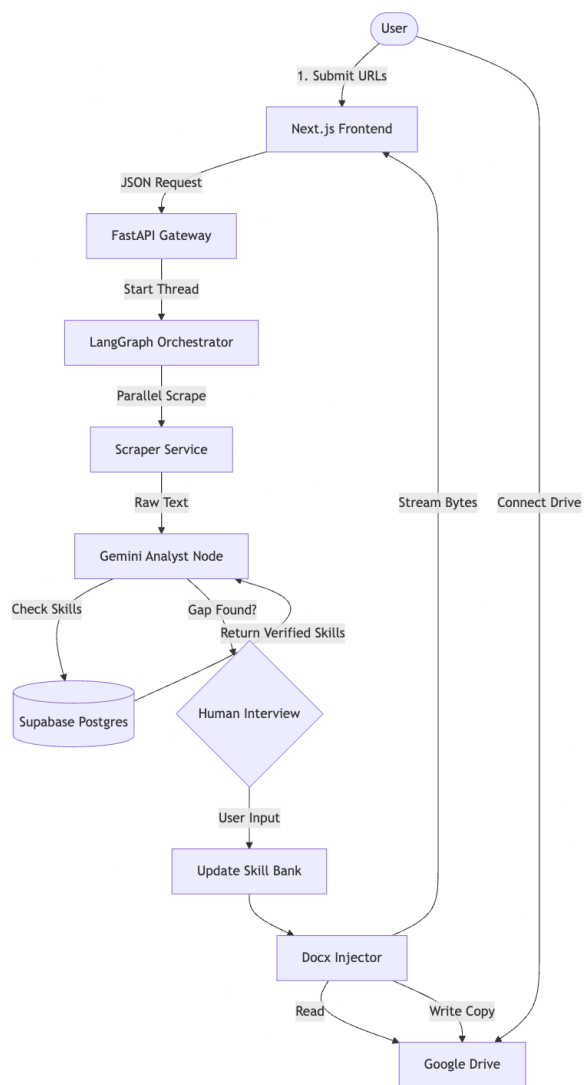
Date: January 2026

Status: Approved for Development

1. High-Level Architecture Diagram

The following flow describes the 'Batch Triage' and 'Resume Factory' logic:





2. Component Interactions & Data Flow

2.1 Phase A: Batch Triage (The "Scan")

1. User submits 3-5 Job URLs via Frontend.
2. API Gateway creates a 'Session' in Supabase.
3. LangGraph spawns parallel 'Scraper Nodes' (using proxies).
4. 'Analyst Node' computes a Feasibility Score (0-100%) using Gemini Flash.
5. Frontend receives a 'Traffic Light' status (Green/Yellow/Red).

2.2 Phase B: Unified Interview (The "Bridge")

1. System calculates the UNION of missing skills across selected jobs.
2. Agent queries 'User Skill Bank' table in Supabase.
 - IF found: Auto-verifies the skill.
 - IF missing: Adds to 'Interview Queue'.
3. User answers questions once via Chat UI.
4. New skills are saved to Skill Bank for future use.

2.3 Phase C: Generation (The "Factory")

1. 'Injector Node' reads the Master Resume (Local bytes or GDrive ID).
2. Node creates a copy: 'Resume-[Company]-[Role].docx'.
3. Node injects keywords into specific paragraph runs (preserving formatting).
4. File is streamed to user (Local) or saved to Drive (Cloud).

3. Storage Strategy (Privacy First)

Data Type	Storage Location	Retention Policy
Session State	Supabase (Postgres)	Persisted (Audit Log)
Resume Files (Binary)	RAM / Google Drive	Ephemeral / User Owned
User Skills	Supabase (Skill Bank)	Permanent (Living Profile)
Scraped Job Text	Redis / Cache Table	24 Hours (TTL)

